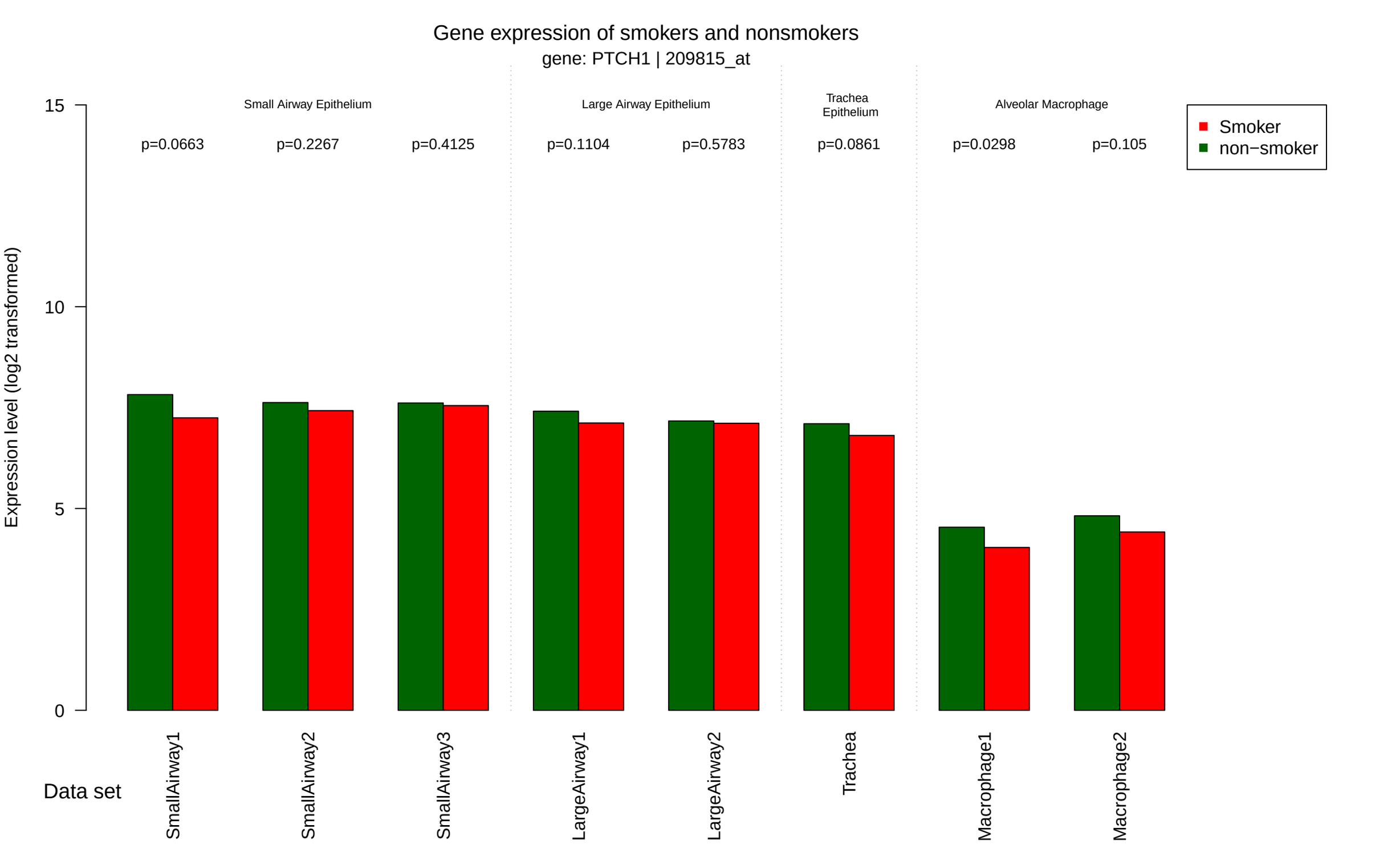
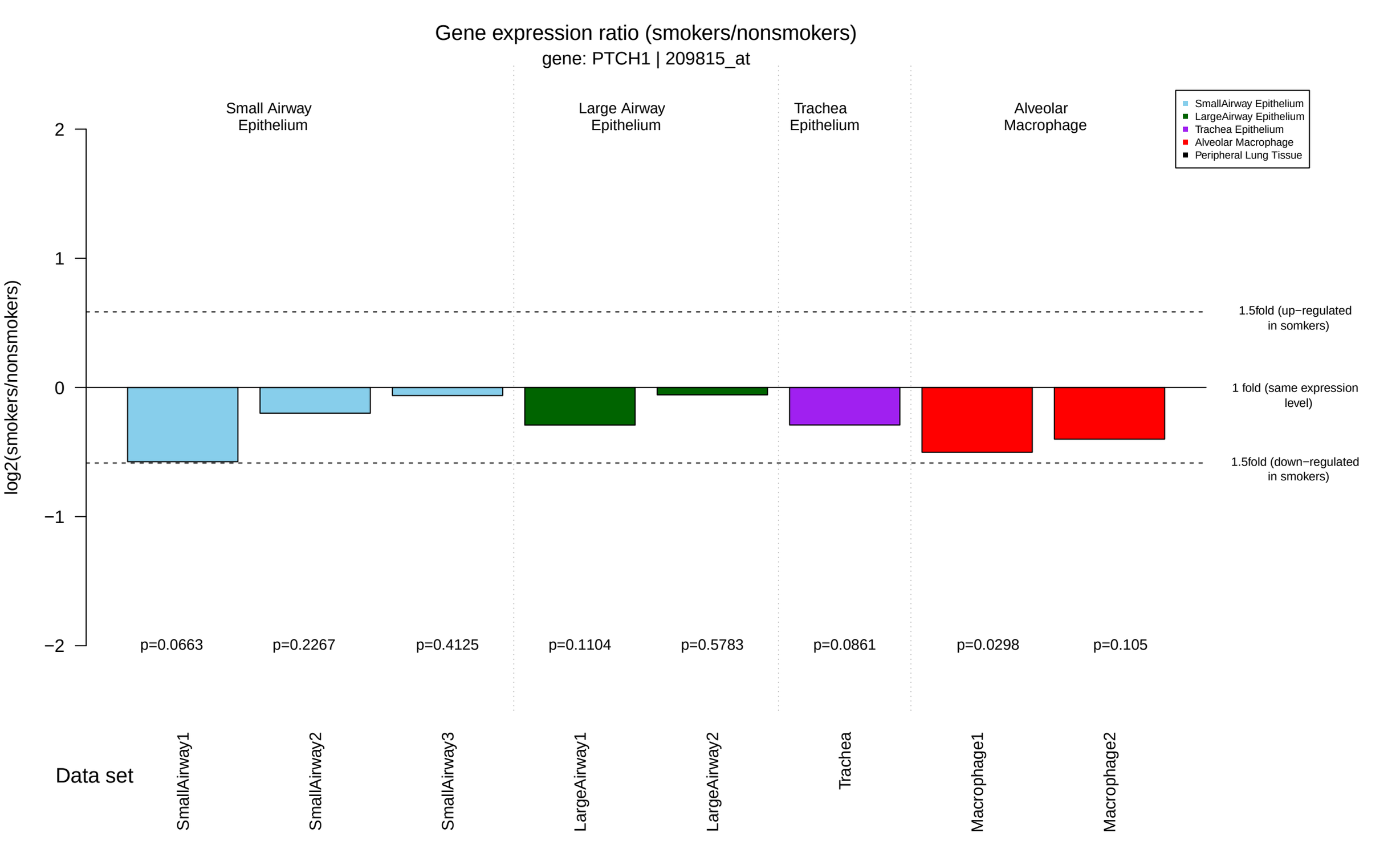
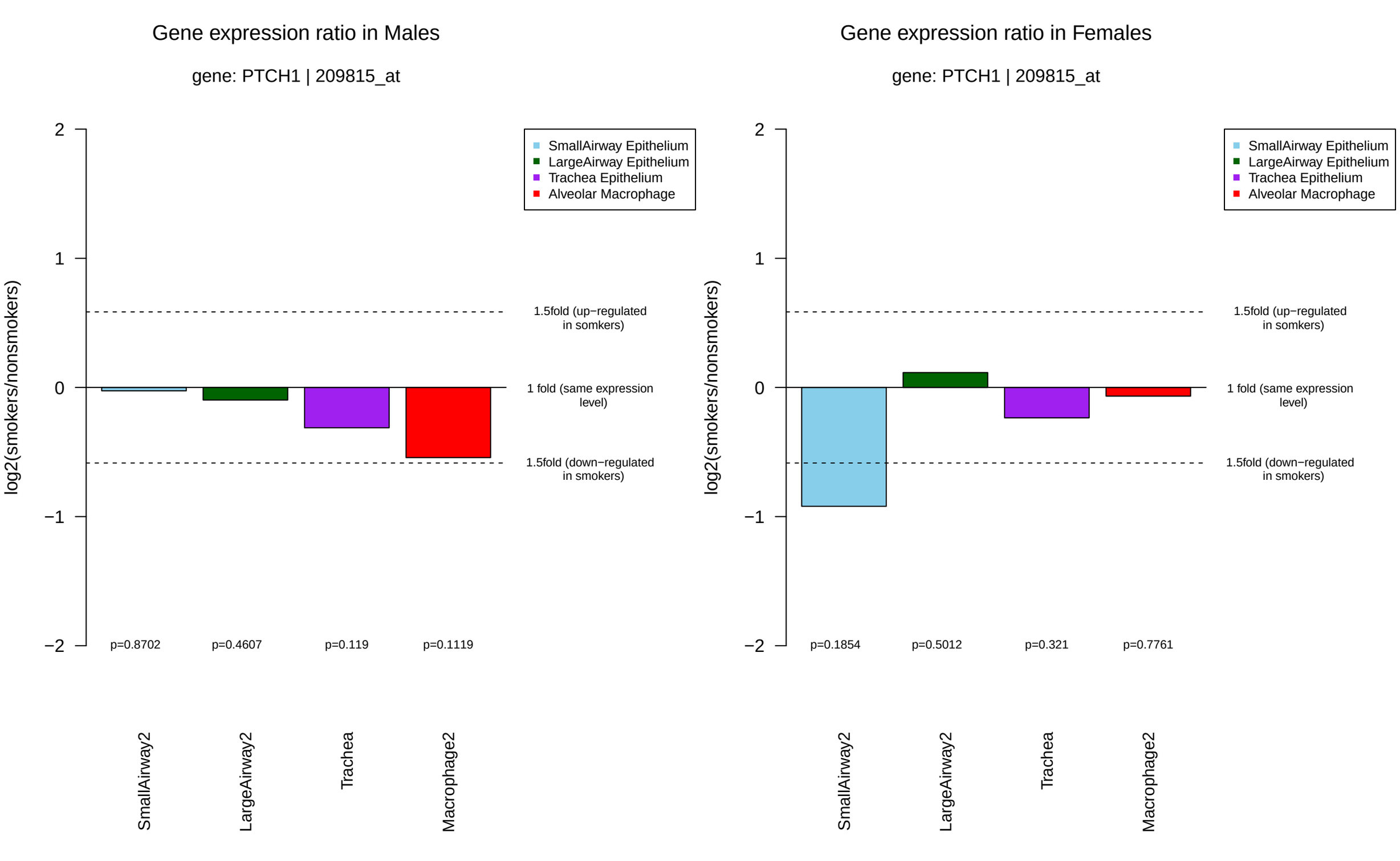
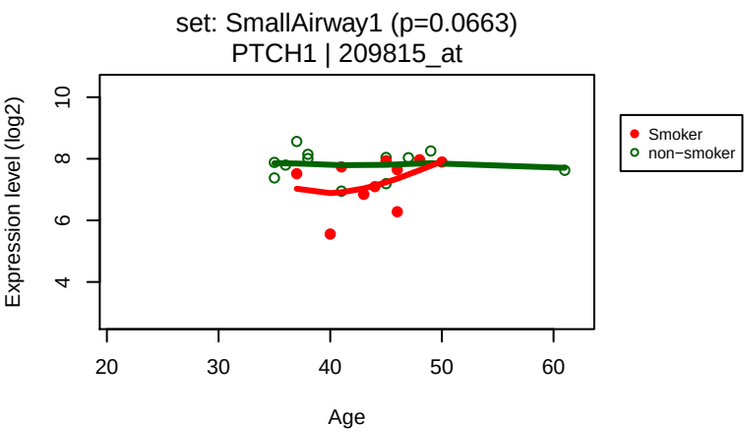
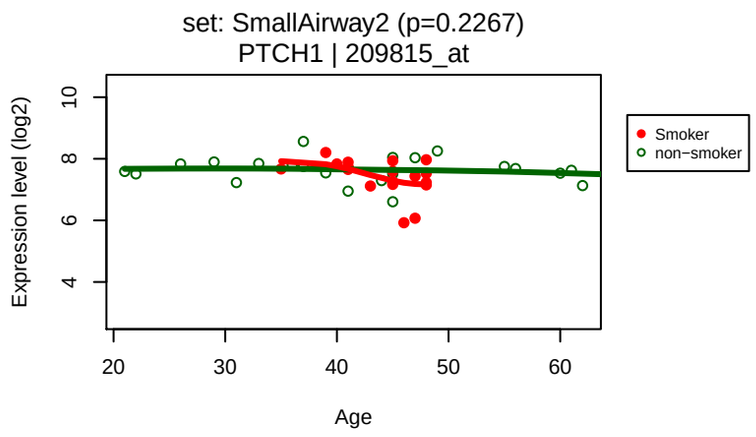
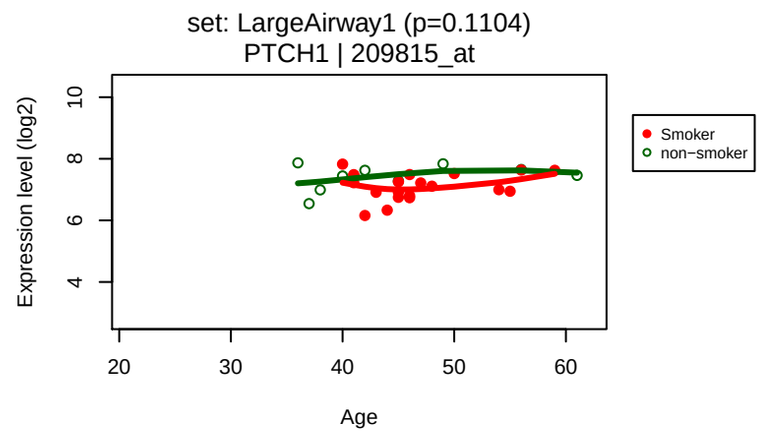

Gene expression of smokers and nonsmokers
gene: PTCH1 | 209815_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0663
p=0.2267
p=0.4125
p=0.1104
p=0.5783
p=0.0861
p=0.0298
p=0.105
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

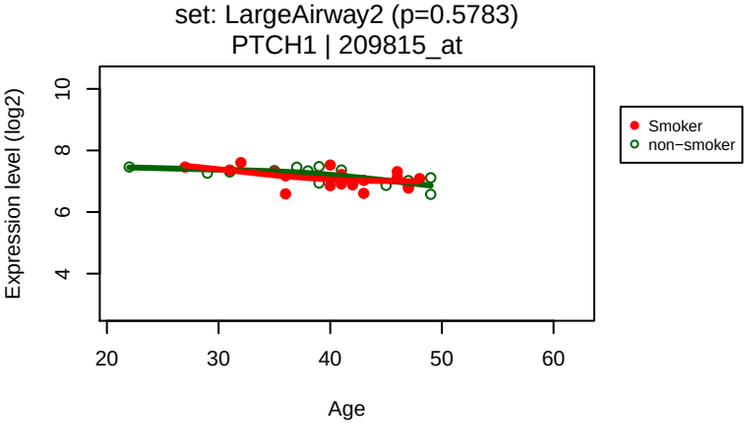

Gene expression ratio (smokers/nonsmokers)
gene: PTCH1 | 209815_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0663
p=0.2267
p=0.4125
p=0.1104
p=0.5783
p=0.0861
p=0.0298
p=0.105
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

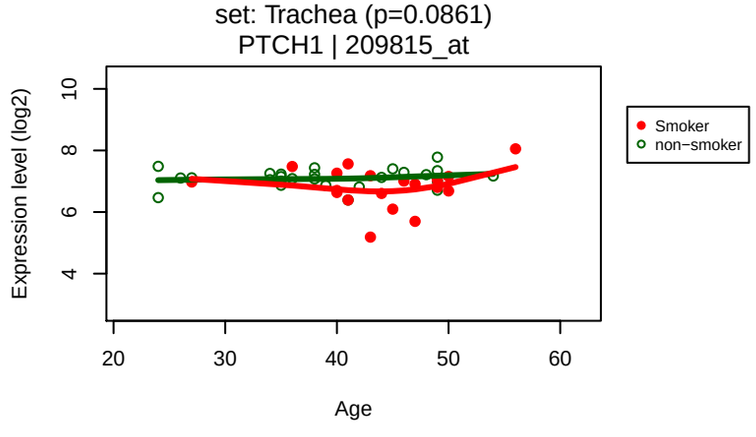

Gene expression ratio in Males
gene: PTCH1 | 209815_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8702
p=0.4607
p=0.119
p=0.1119
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: PTCH1 | 209815_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1854
p=0.5012
p=0.321
p=0.7761
SmallAirway2
LargeAirway2
Trachea
Macrophage2

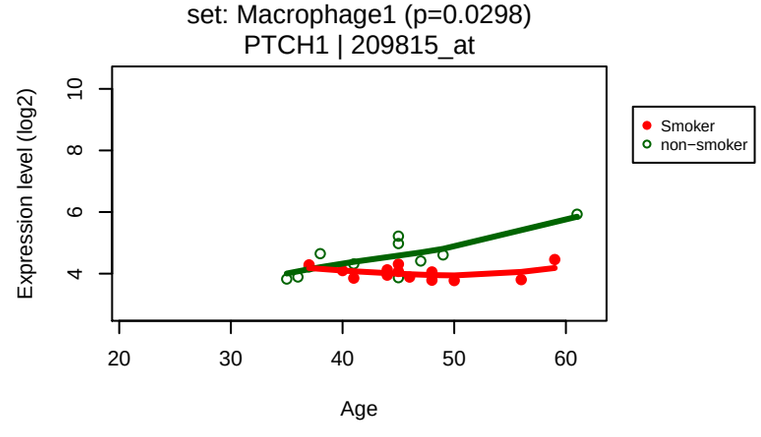

set: SmallAirway1 (p=0.0663)
PTCH1 | 209815_at
Expression level (log2)
Age
Smoker
non−smoker

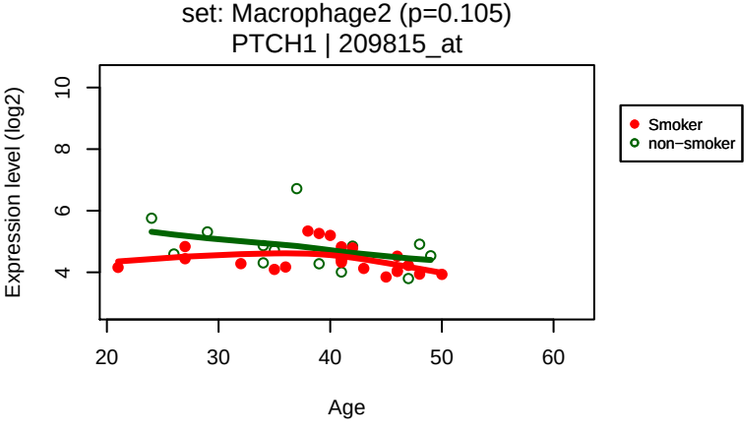

set: SmallAirway2 (p=0.2267)
PTCH1 | 209815_at
Expression level (log2)
Age
Smoker
non−smoker

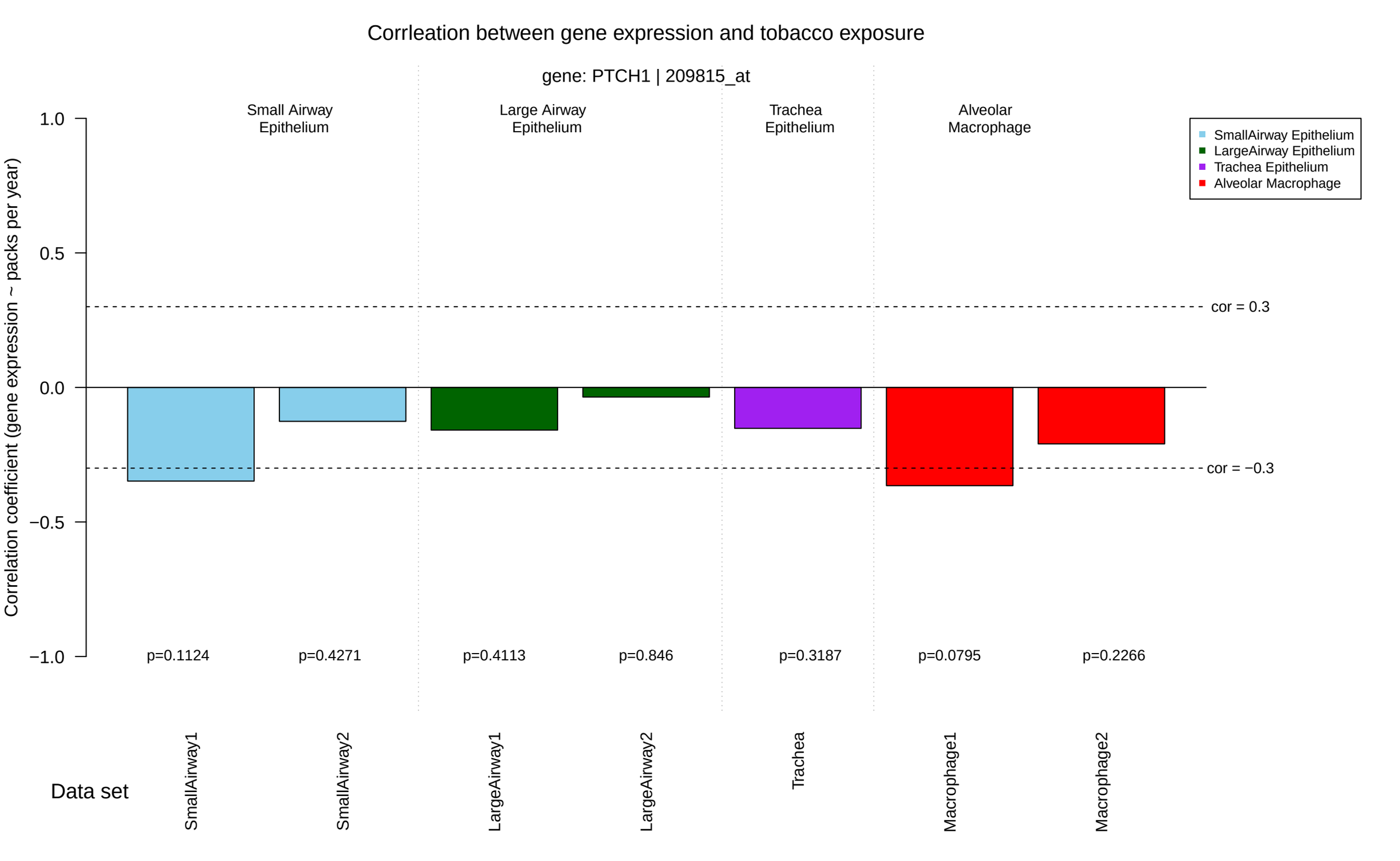

set: LargeAirway1 (p=0.1104)
PTCH1 | 209815_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.5783)
PTCH1 | 209815_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0861)
PTCH1 | 209815_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0298)
PTCH1 | 209815_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.105)
PTCH1 | 209815_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: PTCH1 | 209815_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.1124
p=0.4271
p=0.4113
p=0.846
p=0.3187
p=0.0795
p=0.2266
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2